

Route-Aware Dynamic Traffic Graphs for ETA: Dataset Generation and Interactive Model Testing

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Abstract. Estimated time of arrival (ETA) is the prediction of how long a trip will take from a given origin to a destination under current and evolving traffic conditions. It underpins navigation, dispatch, logistics, and traveler information systems: small errors propagate into poor routing, missed connections, and loss of trust in the service.

Machine learning approaches to ETA require datasets that align road topology, vehicle motion, and labels in a form models can learn from. We present a toolchain that produces *rich, feature-heavy dynamic graph* data suitable for models that use explicit graph structure—for example, architectures with separate treatment of static network elements and time-varying traffic.

The system supports two complementary paths. First, *simulation-based* datasets: the user supplies a SUMO network and simulation configuration; the tool materializes spatio-temporal graph snapshots with detailed per-node and per-edge features from microsimulation. Second, *trajectory-based* datasets: real vehicle trajectories are imported and aligned to the same graph representation via map matching and step conversion on a user-selected region. Porto taxi trajectories illustrate real-world conversion in our bundle; other CSVs and regions are supported.

To demonstrate the value of this representation for learning and comparison, we also built a *web-based testing environment* that couples the simulator to interactive trips, model inference, and logged prediction error, so ETA models can be evaluated under controlled, repeatable conditions.

Keywords: Estimated time of arrival · Traffic simulation · SUMO · Dynamic graphs · Dataset generation · Graph neural networks · Web-based evaluation.

1 Introduction

Estimated time of arrival (ETA) is a central prediction task in transportation systems, informing navigation, dispatch, logistics, and traveler information services. In such settings, ETA is not merely a descriptive quantity: it shapes route choice, scheduling, service reliability, and user trust. For operational platforms, inaccurate or unstable ETA estimates propagate into missed coordination, inefficient routing, and degraded service quality [4].

The central difficulty addressed in this work is not the absence of traffic data in general, but the lack of open, reusable, route-aware data representations tailored to ETA-oriented tasks. Much of the traffic prediction literature relies on sensor-centric or aggregate spatio-temporal datasets, while trajectory-based data often captures movement without preserving the route-aware structure characteristic of navigation-style ETA settings [5]. This problem is compounded by the scarcity of openly reusable ETA datasets: many operationally valuable sources remain proprietary, making reproducible experimentation difficult. The gap is therefore twofold: ETA-oriented learning requires data in which road topology, traffic evolution, and route context are aligned in one coherent form, and that representation should support controlled downstream validation rather than ending at offline dataset export. These requirements become harder still when one seeks to accommodate both simulation-derived traffic data and real trajectory data within a common framework.

This paper addresses that gap through a common ETA-oriented representation constructed from two fundamentally different data sources. The first path begins with controlled simulation, where the evolving traffic state and planned routes are directly observable. The second begins with real trajectories, where route inference and data repair are required before the data can be expressed in a coherent downstream form. Although these paths do not produce the same dataset, they produce distinct dataset instances expressed in the same graph-based representation. Roads are modeled as edges, while junctions and vehicles are modeled as nodes. The representation is explicitly route-aware: vehicles are tracked along known or inferred planned routes, making the data substantially closer to navigation-system usage than to generic traffic datasets. It is also dynamic in two senses: traffic evolution appears in time-varying node and edge features, and the graph structure itself changes as vehicles move and dynamic traffic relations are created and removed. While designed primarily for vehicle-level ETA-oriented tasks, the same representation can also support other downstream objectives, such as road-level load estimation or area-level traffic analysis.

The concrete contribution is an integrated artifact ecosystem composed of a desktop dataset-construction tool and a web-based testing environment. The trajectory path is illustrated on the Porto taxi dataset (ECML/PKDD 2015 challenge; July 2013–June 2014) [12] as a bundled conversion *demonstrator*; evaluation is not limited to that source, and the tools accept other trajectories and geographies. The desktop tool supports both data-construction paths: for simulation-based datasets, an interactive environment for configuring scenario behavior, controlling simulation settings, visually validating zones, landmarks, and vehicle classes, and exporting route-aware graph snapshots with adjustable temporal granularity; for real-trajectory datasets, network selection, trajectory repair, route alignment, and conversion into the same ETA-oriented representation. The web environment exercises the produced representation in controlled downstream use: it supports attachment of a representation-compatible ETA predictor, interactive trip-based testing, registration of prediction outcomes in a database, statistical analysis, plots, and export of PDF reports. As an open-

source evaluation framework, it can be reused with other compatible predictors and configurations. Artifact URLs are omitted for double-blind review and can be restored in the camera-ready version.

The predictive model used within the workflow is not the principal contribution of this article. It serves as a representation-compatible validation component through which the usability of the produced datasets can be assessed. Evidence in Section 6 should be read as support for representation coherence, trainability, and end-to-end workflow functionality, not as a claim of predictive-model novelty or superiority. The attached predictor is a practical downstream instrument for exercising the representation under controlled conditions; the web environment provides evidence that the representation is operational enough for navigation-style ETA testing in an interactive setting.

The remainder of the paper positions the work (Section 2), presents the system overview (Section 3), the dataset-construction paths and shared representation (Section 4), the web-based evaluation loop (Section 5), and an evaluation snapshot with dataset statistics, metrics, and figures (Section 6), followed by discussion and conclusion (Section 7).

2 Related Work and Positioning

2.1 Traffic data, graph models, and ETA

The broader traffic-prediction literature has been shaped by sensor-centric and aggregate spatio-temporal datasets, as reflected in surveys of traffic prediction with big data [5]. Frameworks such as LibCity have made this landscape more reproducible by standardizing datasets, tasks, and model interfaces for urban spatial-temporal prediction [9]. The dominant benchmark orientation remains broad traffic forecasting rather than route-aware ETA estimation in which the planned route of each vehicle is explicit. ETA tasks are inherently path-dependent, whereas many widely used resources primarily capture network state, trajectories, or aggregate flows without preserving a navigation-style route context.

Graph-based learning has become a standard approach to traffic forecasting because it aligns naturally with road-network structure; models such as DCRNN and ST-GCN illustrate explicit graph formulations for spatial and temporal dependencies [7,6]. In the ETA domain, production-scale work on ETA prediction with graph neural networks in Google Maps underscores structured, route-aware modeling for travel-time estimation [8]. Related routing-oriented work explores future-traffic-aware route choice and simulative ETA estimation [10,11]. The present paper differs in emphasis: it focuses on a reusable route-aware representation and the tooling to generate and evaluate it, rather than on a new predictive architecture.

2.2 Simulation, trajectories, and integrated evaluation

Microscopic traffic simulation, with SUMO and TraCI, provides repeatable scenarios and external programmatic control [1,2], and is attractive for ETA-oriented

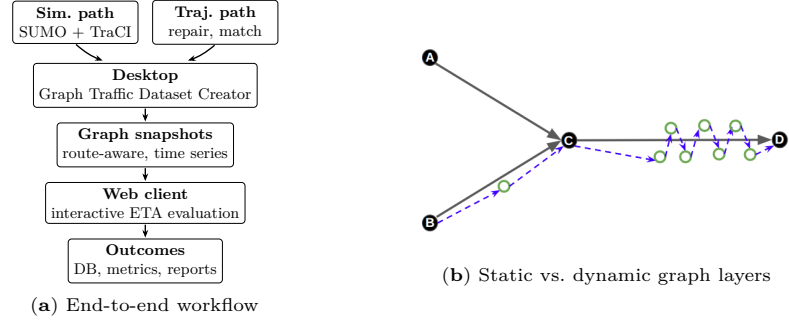


Fig. 1. (a) End-to-end workflow: simulation and trajectory paths merge in the desktop tool, export route-aware graph time series, and feed the web layer for testing and logged analysis. (b) Static layer: black nodes A–D are junctions; grey edges are roads. Dynamic layer: green nodes are vehicles; dashed blue edges connect vehicles to other vehicles and to junctions.

dataset construction because vehicle state, route context, and traffic evolution are highly observable. Real trajectory data requires repair, alignment, and route inference before coherent downstream ETA analysis. Simulation and trajectory pipelines are often optimized separately; this work supports both in one common representation.

Existing libraries and evaluation frameworks improve reproducibility for traffic prediction but typically start from pre-existing datasets rather than from integrated dataset construction. The web-based component here is a downstream evaluation layer coupled to dataset generation: it attaches a compatible predictor, registers outcomes, and exposes analysis and reports under controlled conditions. The paper should be read as an integrated workflow for constructing route-aware ETA-oriented datasets and validating their downstream usability, not as a new ETA-model paper or a generic benchmark library alone.

3 System Overview

The proposed workflow links heterogeneous traffic inputs to downstream ETA-oriented evaluation through a single artifact ecosystem. At a high level, the system begins from one of two data-construction paths: a simulation-derived path built on controlled SUMO scenarios, and a real-trajectory path built on imported GPS-like traces aligned to a road network. Although these paths differ substantially in observability, preprocessing, and error sources, both terminate in a common graph-based representation designed for route-aware ETA tasks. This shared representation can then be exercised through a web-based evaluation environment in which a compatible predictor is attached, trip-level outcomes are registered, and statistical analyses are produced. Figure 1(a) summarizes this pipeline; panel (b) sketches the static and dynamic parts of the graph representation.

Dataset construction is operationalized through the Graph Traffic Dataset Creator, an interactive desktop environment rather than a collection of disconnected scripts. On the simulation side, it supports scenario definition, control of simulation behavior, visual validation of zones, landmarks, and vehicle classes, and export of route-aware graph snapshots with adjustable temporal granularity, including configurable noise such as random untracked trips. On the trajectory side, it supports map download, network preparation, trajectory repair, route alignment, and batch export into the common representation.

The conceptual center of the system is the common ETA-oriented representation: roads as edges, junctions and vehicles as nodes, with each vehicle associated with a known or inferred route. Traffic state evolves in node and edge features, and graph relations change with vehicle movement. The representation is aimed at vehicle-level ETA but supports other views such as road-level load or area-level aggregates. The web evaluation client provides the downstream evaluation layer: interactive testing with an attached ETA predictor, outcomes stored in a database, and statistical summaries, plots, and report generation, bridging representation design and measurable use. The predictive model is a deliberately bounded, representation-compatible validation component; compatible predictors can be attached, replaced, and tested in the same environment, including by external users of the open-source release. The cloud-deployed instance discussed later is one demonstrator configuration of the broader workflow.

4 Dataset Construction and Shared Representation

4.1 Goals and construction paths

The representation is designed first for route-aware ETA learning while remaining useful beyond a single supervised target: it places road topology, vehicles, route context, and time-varying traffic state in one coherent form for controlled experimentation and downstream testing. Because it preserves graph semantics rather than only a final ETA label, it can also support related tasks such as road-level load estimation and area-level analysis.

In the simulation-derived path, construction begins from a controlled microscopic environment built with SUMO and managed through the desktop tool and TraCI [1,2]. Planned routes, vehicle identities, network structure, and evolving traffic states are directly observable during execution. The desktop workflow is configurable: simulation behavior, validation of zones, landmarks, and vehicle classes, temporal sampling of snapshots, and optional irregularity such as random untracked trips for robustness experiments. Figure 2(a) shows the simulation-definition stage; Figure 2(b) shows the web client used to evaluate predictors on simulation-backed journeys.

The real-trajectory path starts from noisier traces: GPS inaccuracies, temporal inconsistencies, or jumps may break a coherent path interpretation. Conversion is a staged repair-and-alignment workflow: network preparation, import, repair, route inference or matching, slicing of trajectories with major jumps,

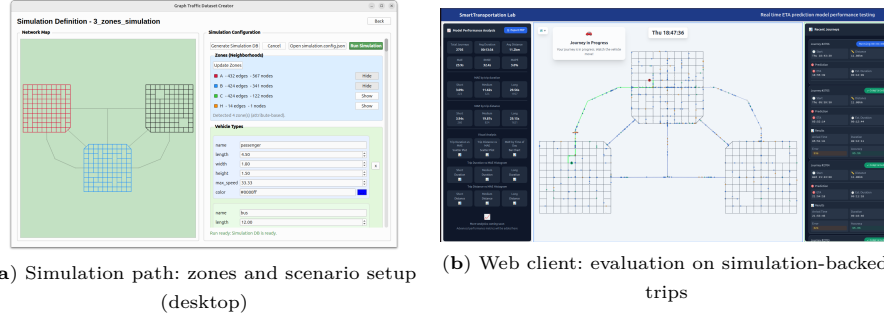


Fig. 2. Simulation construction path and its web evaluation loop. (a) Simulation Definition: SUMO network map with attribute-based zones (A/B/C/H), zone statistics, and configuration before database generation and runs (example project). (b) Real-time ETA testing: model performance summaries, interactive map with active route, and journey log (demonstrator layout; neutralized for review).

and filtering of degenerate origin–destination pairs. We use the Porto taxi trajectory dataset [12] as a representative real-world conversion example in our open-source bundle; the toolchain applies to other CSVs and regions. Figure 3(a) shows the desktop conversion interface; Figure 3(b) shows web-side evaluation on trajectory-backed runs. Figure 4 visualizes route-matching outputs on the map: search scope, candidates, raw GPS, the chosen SUMO route, and projections onto it.

4.2 Route matching on the network (abstract procedure)

Matching is not run on the full regional road graph: first, an axis-aligned bounding box B is built around the GPS polyline P (optionally padded), and $N_{bb} = (V_{bb}, E_{bb})$ is the subnetwork of N formed from all edges whose geometry intersects B and their incident nodes. GPS polylines are then aligned to N_{bb} in two phases: a *scoring pass* assigns every edge a coarse tier (1, 10, or 1000) from successive segment–edge comparisons, then a *routing pass* searches for a low-cost path that respects those tiers. For each polyline segment, edges whose heading differs from the segment by more than $\pi/9$ are discarded; among survivors, edges are ranked by $r_e = \lambda\delta_e + \frac{1}{2}(d(p_i, e) + d(p_{i+1}, e))$, where δ_e is the acute angle gap in radians, $d(p, e)$ is the distance from point p to the centreline polyline of edge e , and λ fixes the radian-to-metre (map-unit) trade-off. The first k edges become *primary* candidates (tier 1), the next j become *secondary* (tier 10); edges that never qualify as a potential match in any segment keep tier 1000. Whenever an edge appears among the top- k candidates for a segment, its tier is set to 1 outright; secondary (10) updates use min with the current tier so an edge that already has tier 1 is never raised to 10 by a later segment. The routing pass runs A^* exactly k times: each run starts from one edge in Top_1 (the k primaries for the first segment) and *terminates* as soon as the search reaches any of the *last* primaries—any edge in Top_{m-1} for segment (p_{m-1}, p_m) —which form the

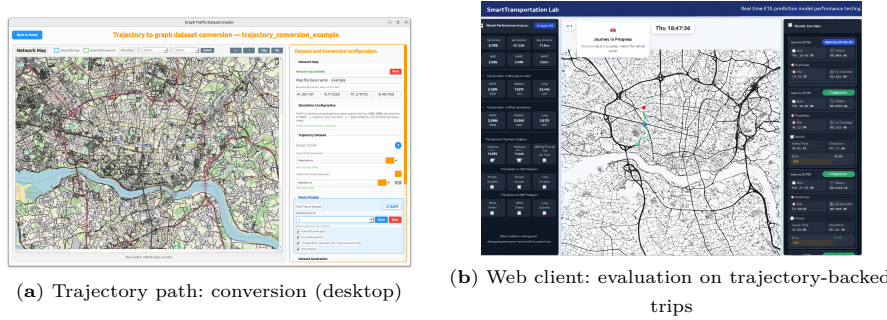


Fig. 3. Trajectory construction path and its web evaluation loop. **(a)** Trajectory dataset conversion: OSM/SUMO map, network overlay, CSV inputs, and batch conversion controls (example Porto-area project). **(b)** Same web client as in Figure 2(b), here exercised on exports aligned with real trajectories: map with active route, aggregate metrics, and recent journeys (demonstrator layout; neutralized for review).

goal set. Edge costs are tier-weighted; the heuristic is Euclidean distance to p_m . The cheapest route among the k runs is returned. Algorithms 1–2 state this procedure formally.

Finally, GPS samples are projected onto the inferred SUMO route so vehicle position can be aggregated as a time series in the shared representation.

4.3 Shared representation, dynamics, and use

Both paths emit dataset instances in the same conceptual form: roads as edges, junctions and vehicles as nodes, with planned or inferred routes so the graph encodes route-conditioned traffic context, not only a traffic snapshot. Exports are feature-rich at node and edge level. Figure 1(b) illustrates static junctions and roads together with vehicle nodes and dynamic edges (vehicle–vehicle and vehicle–junction).

The graph is dynamic in two senses: node and edge attributes evolve with traffic conditions, and structure changes as vehicle nodes move and dynamic relations are created or removed. The stable road graph is supplemented by transient traffic relations reflecting positioning and interaction. Downstream users need not consume every element: they may use the full dynamic graph or simplified views (vehicles, roads, junctions; optional filtering of dynamic relations). The contribution is a reusable representation and workflow, not a single prescribed architecture.

5 Web-Based Evaluation and Validation Loop

5.1 Workflow, interface, and evidence

The web application closes the loop between dataset construction and measurable use. It accepts representation-compatible inputs, runs prediction sessions

Algorithm 1 Edge-weight assignment from a GPS polyline (scoring pass)

Require: Polyline $P = (p_1, \dots, p_m)$; full road network $N = (V, E)$ with directed edges; integers $k, j \geq 1$; angular threshold $\theta_{\max} = \pi/9$; trade-off $\lambda > 0$ (same length units as the map, e.g. $\lambda = 5$).

Ensure: Each edge $e \in E_{\text{bb}}$ has a tier $w(e) \in \{1, 10, 1000\}$ (lower is more attractive).

- 1: Build axis-aligned bounding box B containing all points of P (optional margin).
- 2: Let $N_{\text{bb}} = (V_{\text{bb}}, E_{\text{bb}}) \subseteq N$ be the subnetwork of all edges in E whose centreline intersects B , together with their endpoint nodes in V .
- 3: Initialise $w(e) \leftarrow 1000$ for all $e \in E_{\text{bb}}$.
- 4: **for** each consecutive segment $s_i = (p_i, p_{i+1})$, $i = 1, \dots, m-1$ **do**
- 5: $\phi_i \leftarrow \text{atan2}(p_{i+1} - p_i)$ $\triangleright$ segment heading in the projected map
- 6: $\mathcal{C} \leftarrow \emptyset$ $\triangleright$ candidate edges for this segment
- 7: **for** each edge $e \in E_{\text{bb}}$ **do**
- 8: $\psi_e \leftarrow$ heading of e from its centreline polyline (first to last shape point)
- 9: $\delta_e \leftarrow \min(|\phi_i - \psi_e|, 2\pi - |\phi_i - \psi_e|)$
- 10: **if** $\delta_e \leq \theta_{\max}$ **then**
- 11: mark e as *potential* for s_i ; add e to $\mathcal{C}$
- 12: $d_i \leftarrow d(p_i, e)$; $d_{i+1} \leftarrow d(p_{i+1}, e)$ $\triangleright$ shortest Euclidean distance from each segment endpoint to the polyline of e
- 13: $r_e \leftarrow \lambda \delta_e + \frac{1}{2}(d_i + d_{i+1})$ $\triangleright \lambda$ scales the angle gap δ_e (radians) into map units so it is comparable to the mean endpoint–edge distances.
- 14: **end if**
- 15: **end for**
- 16: sort $\mathcal{C}$ by increasing r_e
- 17: $\text{Top}_i \leftarrow$ first k edges; $\text{Sec}_i \leftarrow$ next j edges (if $|\mathcal{C}| < k+j$, take as many as exist)
- 18: **for** $e \in \text{Top}_i$ **do**
- 19: $w(e) \leftarrow 1$
- 20: **end for**
- 21: **for** $e \in \text{Sec}_i$ **do**
- 22: $w(e) \leftarrow \min(w(e), 10)$
- 23: **end for**
- 24: **end for** $\triangleright$ Edges that were never marked potential in any segment keep $w(e) = 1000$. Each time an edge ranks in Top_i , set $w(e) \leftarrow 1$; secondary ranks use min with 10 so a prior tier 1 is not overwritten by a weaker segment.

Algorithm 2 Route extraction by multi-start A*

Require: Restricted network N_{bb} ; tiers $w(e)$ on E_{bb} ; sets Top_1 and Top_{m-1} from Algorithm 1 (primaries for the first and last polyline segments); last point p_m .

Ensure: Shortest-cost route among k multi-start runs.

- 1: Goal set $G \leftarrow \text{Top}_{m-1}$ $\triangleright$ the k primary edges for the last segment (p_{m-1}, p_m)
- 2: **for** each start edge $e_{\text{start}} \in \text{Top}_1$ **do**
- 3: Run A* from e_{start} with edge cost $c(e) = w(e) \cdot \text{len}(e)$ and $h(v) =$ Euclidean distance from v to p_m ; **stop** the run as soon as the search reaches any state on an edge in G .
- 4: **end for**
- 5: **return** the route with minimum total cost among these k runs.

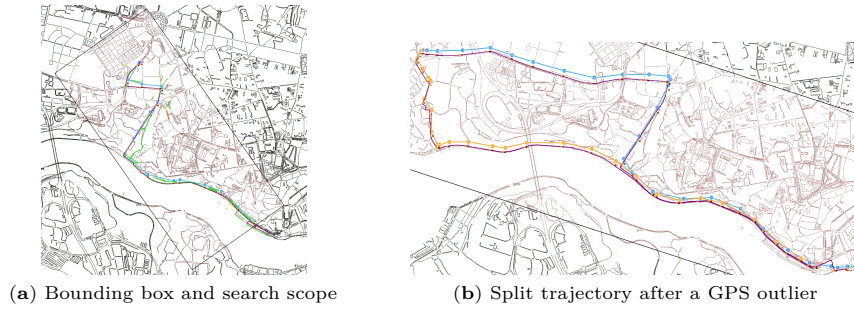


Fig. 4. Route matching on the imported SUMO network (conversion path). Panels (a) and (b) both show the *bounding box*, *burgundy* roads and junctions that limit the matcher’s search entities, and the thick *dark* final route on the SUMO network. Panel (a) additionally shows the raw GPS trace as a *light blue* polyline (numbered blue samples) and *green* segments as the top-three candidate edges at sampled locations. Panel (b) additionally shows *light blue* and *orange* polylines for the *split* trajectory—two segment polylines after cutting at an outlying GPS point (e.g. ~ 5 km from adjacent samples), with each segment matched separately.

under controlled conditions, and persists evidence for later inspection, so the representation is exercised end-to-end rather than left as a static export. The environment supports interactive ETA-oriented testing, cumulative storage, and repeatable analysis; it is not a production navigation system, but it shows navigation-style interaction is feasible on the representation. Section 4 places the simulation path first: Figure 2(b) shows the main evaluation dashboard for simulation-backed trips (map, cohort metrics, journey log). The trajectory path uses the same client after export, as in Figure 3(b), following desktop conversion in Figure 3(a).

A compatible ETA predictor is attached as a pluggable component that consumes the shared representation and returns trip-level ETA outputs; the design is representation-centered rather than architecture-centered. Users can trigger predictions for selected trips, inspect outputs, and run controlled sessions; other predictors can be substituted if they conform to the representation interface. Predictions and metadata are stored in a database so results can be revisited and aggregated. The client computes measures such as MAE, RMSE, and MAPE over selected sets and supports cohort grouping. It also generates plots from registered data and can assemble PDF reports combining statistics and figures.

5.2 Demonstrator scope and validation role

The deployed application is one configuration of a broader framework: the same logic can be reused with alternative predictors and datasets from supported environments. The contribution is the coupling of construction and evaluation through a reusable interface, not a single prototype deployment.

The web loop is included for methodological validation: representation coherence, trainability and testability downstream, and a workflow that does not stop

Table 1. Simulation vs. trajectory conversion (representative bundles in the open-source examples). Junction/road-edge counts from SUMO `net.xml` (non-internal edges only). Coverage from `location` bounding box. Simulation snapshots assume dense sampling every Δt over four weeks; trajectory snapshot count is the *actual* number of emitted graph steps (temporal sparsity in the trace), not dense $\lfloor T/\Delta t \rfloor$.

Quantity	Simulation	Trajectory conversion
Temporal span	4 weeks	1 year
Project name	3 zones city	Porto taxi
Trips after repair	403 848	1 312 637
Junction nodes	1 031	75 264
Road edges (non-internal)	1 294	146 816
Approx. coverage area (km ²)	≈ 203	≈ 385
Graph snapshots (at Δt)	80 640	925 644
Snapshot period Δt (s)	30	30
Node feat. dim. / edge feat. dim.	28 / 6	28 / 6

Table 2. ETA error on held-out evaluation journeys: route-aware model vs. mean-duration baseline (train-set mean predicted for every test trip). *Simulation* = graph data from the microsimulation path; *trajectory conversion* = real traces aligned to the network. MAE/RMSE in seconds; MAPE in percent.

Data source	n	Route-aware model			Mean baseline (avg)		
		MAE (s)	RMSE (s)	MAPE (%)	MAE (s)	RMSE (s)	MAPE (%)
Simulation	2 707	46.16	107.15	17.99	260.63	325.13	127.75
Trajectory conversion	3 055	52.38	121.50	20.41	282.66	352.41	138.50

at raw export. Evidence from the web client should not be read as a claim of model novelty; it supports usability of the representation and continuity between construction and evaluation.

6 Evaluation

7 Discussion and Conclusion

The main contribution of this paper is an integrated workflow for constructing route-aware ETA-oriented graph datasets and validating their downstream usability through a connected web evaluation environment. The paper has therefore been organized around artifact integration rather than around predictive-model novelty. The desktop tool addresses the front half of the problem by supporting dataset construction from both controlled simulation and repaired real trajectories, while the web environment addresses the back half by turning those outputs into persistent, analyzable evaluation evidence. Taken together,

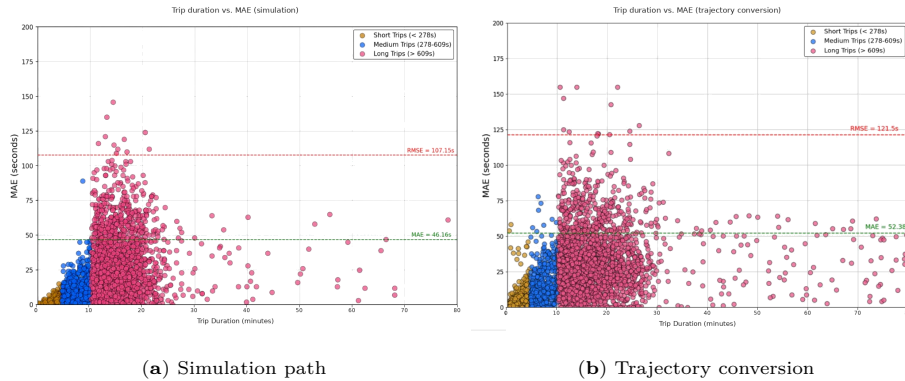


Fig. 5. Trip duration vs. absolute ETA error (MAE in seconds) on held-out evaluation journeys. Horizontal lines show aggregate model MAE (green) and RMSE (red) for the route-aware predictor. (a) Microsimulation-backed graph data. (b) Real trajectories after map matching and conversion.

these components reduce the gap between dataset preparation and downstream ETA-oriented experimentation.

This framing also clarifies the role of the attached predictor. It is included as a validation component, not as the paper’s primary contribution. Its importance lies in showing that the shared representation is coherent enough to support downstream ingestion, interactive testing, and measurable analysis across both construction routes. In this sense, the reported evidence should be read as confirmation of workflow usability and representation continuity rather than as a benchmark-style claim about model superiority.

At the same time, the present demonstrator has clear limitations. The deployed web instance represents one practical configuration of the broader framework, not the full space of possible deployments, networks, or predictors. The real-trajectory branch also remains inherently constrained by source-data noise, route-inference uncertainty, and data-availability limits that do not arise in the same way under controlled simulation. More broadly, the current paper emphasizes workflow coherence and proof of usability over exhaustive comparative experimentation.

These limitations point directly to future work. Natural extensions include attaching alternative compatible predictors, evaluating the framework on additional road networks and broader trajectory sources, and expanding the scale and diversity of the reported experiments. Another promising direction is co-operation or integration with navigation applications for live or near-real-time data testing, which could strengthen the practical relevance of the web evaluation loop while remaining consistent with the paper’s representation-centered design. Under that broader perspective, the proposed toolchain is best understood as a reusable foundation for route-aware traffic-dataset construction and controlled ETA-oriented research.

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